Towards Swarm Intelligence Architectural Patterns – an IoT-Big Data-AI-Blockchain convergence perspective

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ABSTRACT

The Internet of Things (IoT) is exploding. It is made up of billions of smart devices -from minuscule chips to mammoth machines - that use wireless technology to talk to each other (and to us). IoT infrastructures can vary from instrumented connected devices providing data externally to smart, and autonomous systems. To accompany data explosion resulting, among others, from IoT, Big data analytics processes examine large data sets to uncover hidden patterns, unknown correlations between collected events, either at a very technical level (incident/anomaly detection, predictive maintenance) or at business level (customer preferences, market trends, revenue opportunities) to provide improved operational efficiency, better customer service, competitive advantages over rival organizations, etc. In order to capitalize business value of the data generated by IoT sensors, IoT, Big Data Analytics/IA need to meet in the middle. One critical use case for IoT is to warn organizations when a product or service is at risk. The aim of this paper is to present a first proposal of IoT-Big Data-IA architectural patterns catalogues with a Blockchain implementation perspective in seek of design methodologies artifacts.

keywords : IoT, AI, Big Data analytics, patterns, decision making, swarm intelligence.

1. INTRODUCTION

All agree on the fact that IoT field is hot, but is still in its infant age, it lacks from methodologies and design reference models. Among many hot research questions in Iot-Big Data integration (security, protocol heterogeneity, time usefulness, data provenance, data governance, etc. [11]). Being able to collect and process IoT data within Big Data framework will deliver very high impact. However, this IoT-Big Data is still achieved on a ad-hoc style depending either on data, or technologies. The aim of this research is to assist IoT-Big Data strategic integration field with integration pattern catalogue to keep the integration architect mission easier and on time. The result may also help solution editor to be compliant between each other and to insure thus more interoperability which is the key word for enhancing added value and insuring a higher return on investment for businesses. The rest of this paper is organized as follows: at first a literature overview of our concept of interest is given in Section 2 followed by our research question in Section 3. In Section 4, the research methodology will be given. In Section 5, 6 and 7, the result catalogue of patterns will is introduced. A conclusion and bibliography will end the paper.

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2. LITERATURE OVERVIEW

2.1 IoT, Data and Artificial Intelligence

The Internet of Things (IoT) was first used by Kevin Ashton in 1999 [3], being a collaboration of a huge number of small sensors and devices in a very interesting end-to-end communication structure. Years later, Blockchain made its entrance as the technology of the future, introducing the concept of distributed platform for financial transactions. This same platform had and still makes a bigger impact on all businesses, on all aspects of the digital life. If we usually bring IoT and blockchain together, it's because their alliance had a great technological potential. If we speak about distributed systems, it is because many research projects funded by various governments [3] are studying the need to 1) locate the intelligence and the provisioning of service at the edge of the network, 2) make the different entities in an IoT architecture collaborate to reach a common goal, so that each entity can make a decision by itself, and by this decreasing the need to relegate the decision making to higher entities in the architecture. These two principals, edge intelligence and collaboration, are the key elements to explain the difference between IoT approaches. While Big Data applications are characterized by having a particular challenge in one or more of the 3 Vs: Volume, Velocity and Variety, IoT projects present challenges in combinations of many V's. Most challenging IoT applications impact both Velocity & Volume and sometimes also Variety [18]. Also, Big Data is getting bigger largely thanks to IoT [25]. The connection of IoT and Big Data can offer: i) deep understanding of the context and situation; ii) real-time actionable insight;

iii) performance optimization; and iv) proactive and predictive knowledge [6]. According to [4], the intersection of IoT and Big Data is creating a tremendous business opportunity, but the reality of today's IoT projects is that this potential has yet to be fully tapped. In fact, IoT and Big Data have stood apart from each other disabling their big potential. Also, integrating diverse data types is in fact a complex task in merging different systems or applications [5]. Successful integration refers to having a uniform view of different data formats providing a single view of the data arriving from different sources and combining the view of data. For helping reducing big data heterogeneity and thus supporting IoT - Big Data integration, [11] proposes a taxonomy of Big Data and Analytics Solutions for IoT Systems, while, [21] proposes a framework for real time semantic annotation of streaming IoT data (Adding Semantics to IoT Data).

2.2 Block Chain & Smart contracts

Blockchain gained enormous interest in and beyond tech industry. It guarantees a set of characteristics to the transactions running on it: transparent, highly resistant to modification, secure and auditable. The technology is evolving to a revolution of the industry business model. However, a global commercialization is a work in progress [22]. The consensus protocols guarantee that all the transactions occurring in the network are trustworthy and that all the participating nodes in the public shared ledger(database) use a secure, real-time, efficient, and reliable agreement mechanism. The consensus mechanism can be categorized based on the types of blockchain i.e. public or private blockchain. In public blockchain, Proof of Work (PoW) [24] and Proof of Stake (PoS) [20] are the most commonly used consensus algorithms. When addressing AI applications, some protocols seem to be more relevant than others. AI applications have high frequency algorithms and need to write operations and continuously update the decision structures for informed decisions. In this case, PoW protocols may become performance bottleneck. In addition, 51% attack on the underlying network can compromise the security of AI applications. PoS could be useful for delay-tolerant AI applications and proves to be energy-efficient, but PoS based protocols are not suitable when AI applications need to handle streaming data, detect changes, and performed real-time informed decisions.

According to the Harvard Business Review [15], smart contracts are the one blockchain application worth investing in. in the same review, Marco Lansiti and Karim R.Lakhani reflected on the time needed for blockchain to reach its full potential and integrate and transform our digital world. If TCP/IP protocols took years to be fully and widely implemented, since they emerge to the surface in 1972, blockchain will need decades. The study defined two key parameters that help us predict the impact of a technology through time: 1) the level of complexity and coordination of the technology, 2) how new the technology is. In one hand, smart contracts are a feature of a very new technology (blockchain), on the other hand, in order to fully implement smart contracts in any structure, it needs a great deal of coordination between parties, and since it is an automated distributed approach of communication in a network, it is very complex. Based on this, smart contracts are seen as future, powerful feature, worth long-term investments to unleash its potential.

3. RESEARCH QUESTIONS

The current proposal focuses on two main questions related to IoT-Big Data-IA design methodologies and architectural patterns as follows:

- **Q1.** When integrating conceptually heavy dual converging paradigms like IoT-Big Data, there is always a modeling methodology issue. Meet-In-The-Middle methodology consisting of conducting in parallel bottom up and top down approach, with its iterative deductive and inductive sprints, it is loosely coupled creative methodology enabling both reuse/mutualization and disruptive thinking. However, this may yield multilayer hardware, network, and analytics software verified modules and micro-services suffering after integration from global processing redundancy, incoherence, on performance or security issues and "**Which methodology to design and integrate an IoDA Architecture (Figure1)?**". Many sub-questions can then be interesting to study: **Q1.1.** "*What methodologies and models to formalize IoT produced data and metadata integration as input of Bigdata analytics ?*"; **Q1.2.** "*is it possible to model/formalize holistically an IoDA Architecture – through a universal IoT-Big Data reference model ?*"; **Q1.3.** "*What is notation/formalism or what are the best of bread notations/formalism to model/formalize an IoDA Architecture ?*"; **Q1.4.** "*is it possible to validate formally an IoDA Architecture modeling? : e.g. the coherence of analytics logic embedded within different edges, fogs, and cloud layers ?*";etc.

- **Q2.** When integrating IoT processing and Big Data analytics, some obvious recurrent paradigms appear like (i) End-to-End Processing and Analytics Chain of Responsibility Orchestration over Micro Computing elements [30], (ii) Forward and Backward Cascading analytics chain [8], (iii) dynamic service migration over multi-cloud levels (cloud, fog, edge) [6], (iv) Actioning Big Data Insights with operational business processes [16], and (v) BI reporting triggers. A deeper study needs to be conducted in order to answer to the question: "**Which are major Architectural patterns an architect should carry off to build relevant IoT-Big Data Architecture ?**". Many sub-questions can then be interesting to study: **Q2.1.** "*Are IoT multi-cloud levels and inherent paradigms (e.g. Cloud, Fog & Edge Computing), and Big Data architectures (Lambda, Kappa, & Smack architecture) immortal schemes and then IoDA integration patterns will be the result of integration of IoT and Big Data existing patterns or IoDA architecture will yield new integrated ones ?*"; **Q2.2.** "*If Big Data architecture has accelerated innovation on Machine Learning algorithms implementation, IoDA Architecture is creating a new paradigm shift on Machine Learning algorithms. What will be new Machine Learning algorithms implementation patterns with regards to IoDA Architecture ? What will be the impact of dynamic service migration among cloud levels on Machine Learning algorithms implementation dynamic adaptation?*"; **Q2.3.** "*Are the existing languages, computing engines and data communication protocols still up to date with the maturation of IoDA Architecture ?;etc.*"

4. RESEARCH METHODOLOGY : DESIGN SCIENCE METHODOLOGY

Hevner [14] states that the main purpose of design science research approach is achieving knowledge and understanding of a problem domain by building and application of a designed artifact.

This research proposal will follow a design science approach according to seven steps; **Guideline 1**: Design as an Artifact – producing a viable artifact in the form of a construct, a model, a method, or an instantiation. **Guideline 2**: Problem Relevance – developing technology-based solutions to important and relevant business problems. **Guideline 3**: Design Evaluation – demonstrating the utility, quality, and efficacy of a design artifact via well-executed evaluation methods. **Guideline 4**: Research Contributions – providing clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies. **Guideline 5**: Research Rigor – applying rigorous methods in both the construction and evaluation of the design artifact. **Guideline 6**: Design as a Search Process – utilizing available means to reach desired ends while satisfying laws in the problem environment. **Guideline 7**: Communication of Research – presenting to both technology-oriented as well as management-oriented audiences. Our research approach will be conducted through a literature Review to identify the research gap and deepen the research questions previously mentioned before tackling the seven guidelines. The patterns that this paper presents are our design artifact. Our proposal, in the present paper, is limited in the Guideline 1, which indicates the design artifact -a result of a investigative process - within a domain problem context. Guidelines 2 to 7 will be implemented for the application, evaluation and verification of the pattern catalogue which represents our proposal, in our future work.

[26] classifies according to [23] swarm organizations common goals to seven categories: (i) decision-making (bacteria), (ii) self-assembling (cells embryo-genesis), (iii) swarm coordinated aggregation (amoebas molding), (iv) swarm coordinated motion (birds flocking), (v) sorting of items (ants brood sorting), (vi) coordinated construction of complex structures (spiders web weaving, and termites nest building), and (vii) coordinated exploration of the environment (ants foraging). Without losing in generality, we present architectural patterns (design artifacts) in three different categories organization patterns, decision making patterns, and learning flows patterns. Each pattern will be described as instructed in [29], using four sections : **Intent** section answers the following questions: What does the design pattern do?What particular design issue or problem does it addresses?;**Motivation** section illustrates a scenario of a design problem and how the class and object structures in the pattern solve the problem. The scenario will help you understand the more abstract description of the pattern that follows; **Known Uses** section details examples of the pattern found in real systems. And finally, a **Blockchain solution** section will give a description of how a pattern is implemented in a blockchain context.

5. ORGANISATION PATTERNS

5.1 Galaxy Pattern – Decentralised

5.1.1 Intent

Research works in AI have shown that decentralization is a key transformation towards a robust, reliable and fault tolerant system[33]. Decentralization addresses the activity of an autonomous agent in multi-agent systems: a group of intelligent entities, where each entity acts according to its current state of knowledge, and each entity's existence is independent from other entities. The main issue in decentralized systems is to study the structure of the entities to see what tasks they can perform. In decentralized systems, every entity makes its own decision. The behavior of the system is the aggregate of the decisions of the individual entities. There is no single entity that handles a request by its own.

5.1.2 Motivation

Decentralized systems gained enormous interest in and beyond tech industry. It guarantees a set of characteristics to the transactions running on it: transparent, highly resistant to modification, secure and auditable. The technology is evolving to revolutionizing the industry business model. Bitcoin [9] is the most popular use case of decentralized systems. It is an aggregate of all the nodes that mine the cryptocurrency. It uses two principles: 1) transactions are created by the users of the network; 2) each user has a record of all transactions, forged into blocks, within a specific time stamp.

The IoDA Front end Architecture is composed of two parts Device/Sensor and FOG/Edge that insure two types of frontal processing and analytics of incident management.

Connected Device/Sensor Processing & Analytics: insure acquisition of incident data, and incident data filtering and Simple classifier.

Fog/Edge Processing & Analytics : insure incident data processing (Detection of anomaly incident), incident data analytics : Pattern recognition/correlation/scoring (advanced supervised time-based analysis algorithms here need smaller training set but may need more performance resources like GPU), and incident data routing : Transmission of the anomaly information through an **Edge Spooler**.

The IoDA Architecture Back-End is represented by several components insuring cloud side processing and analytics of incident management.

Cloud Data Processing & Analytics : insure incident data routing (Transmission of the anomaly information to the relevant back end processing & analytics system – **ESB/CEP**), incident data analytics and intelligence : Extraction, cleaning and annotation, Integration, aggregation and representation, Modeling and analysis Pattern recognition/correlation/scoring (more sophisticated supervised machine learning algorithms (e.g. deep learning) may here need big training sets) (**Big Data**), incident data processing : Anomaly Human Processes (Human qualification of the anomaly information) and Enterprise Business Processes (**BPMS based on Big Data analytics**), and Interpretation : through on **Reporting incident KPI Scoreboards** based on (**Data Warehousing** & **Data Viz**). One could connect those components to other enterprise business systems ERP, CRM, PLM.

5.1.3 *Known Uses*

Cryptocurrencies, Blockchain, Decentralized databases, Edge Computing and Fog Computing are all known uses of the galaxy pattern.

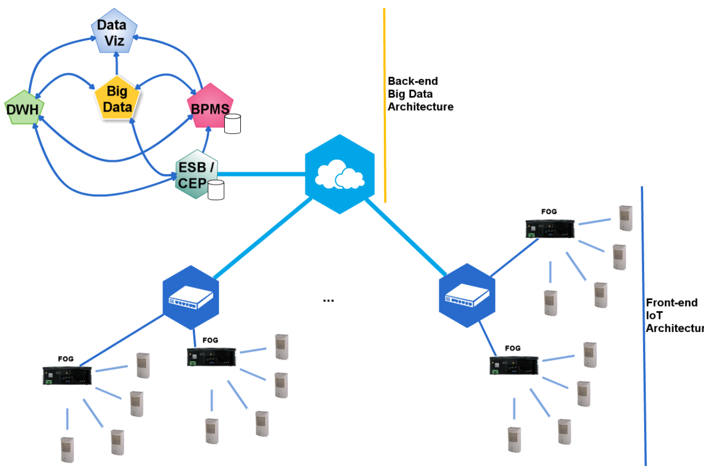

Figure 1: IoDA IoT - Big Data end-to-end Integrated Architecture

5.1.4 *Blockchain solution: Atomic*

One of the biggest issues we can come across is: how mature the exchange between multiple blockchains is? The answer to this question gives us an update on how decentralized cryptocurrency platform are. In fact, they are not as decentralized as we think they are. Imagine the scenario that follows: a swarm of computers, made by the same company, running the same software and exchange the same currency. If all computers happen to run a software with a defect, or if the majority of the coins are held in one transaction, or if the manufacturer decided of some changes on the hardware, these matters may question the political, architectural and logical decentralization of the exchange platform, as the system may act a a single structure. The true decentralization may start at being able to exchange assets regardless of the technology and the currency. Solutions like Atomic [19] and Stellar are blockchains solution that do just that.

5.2 Mesh Pattern – Distributed

5.2.1 *Intent*

The difference between decentralized and distributed is subtle: A decentralized system is a subset of a distributed system. The main difference is how and where the decision is made and how the information is shared through the system. in the distributed case, the processing is shared across multiple nodes. the nodes may vary from simple processing elements to complex ones with rational behavior and important computation capacity but the decisions may still be centralized and use complete system knowledge.

5.2.2 *Motivation*

The distribution pattern refers to a collaborative solution of global problems by a distributed group of entities. The task is initially defined and the issue is to design the distributed entities group to solve the task. Google search system is a fair example of that design: in a user point of view, the browser is a single system that delivers a result to the query. Instead, each query is processed by millions of nodes/computers through the web and return a relevant result.

5.2.3 *Known Uses*

SmartMesh [13] is a blockchain based underlying protocol of the Internet of Things. SmartMesh has built-in blockchain light nodes, and extends the Raiden and Lightning Networks second layer architecture network protocols allowing internet-free digital payments and transactions. Utilizing blockchain-based token incentives, SmartMesh technology allows the formation of agile, decentralized Mesh Networks that can self-repair and have higher near-field speeds and bandwidth than standard Internet connections. SmartMesh allows blockchain to break through the Internet boundary into the IoT (Internet of things) and IoE (Internet of Everything) era where all things are connected.

5.2.4 *Blockchain solution*

The work in [34] proposed a blockchain-based trust approach related to message dissemination in VANET. By disseminating critical event information, VANET or Vehicular Ad-Hoc Network plays a crucial role in securing drivers life and property. The proposal adopts a PoW consensus mechanism, with safety event messages as its asset instead of crypto-coins. [34] represents a blockchain dedicated to that matter: all the vehicles in the network download and update the blockchain. In that scheme, the blockchain stores all the history of vehicles trust level along with event message. When an accident happens, the vehicle that encounters the event broadcasts the event message to neighbor vehicles in the blockchain networks. When a vehicle receives a new event message, they verify the location certificate embedded in the event message if it belongs to the same area indicated by the certificate. Then, the neighbor vehicles check other parameters of the event messages. Every vehicle independently checks each event message before broadcasting it further.

6. DECISION MAKING PATTERNS

6.1 Consensus Pattern

6.1.1 *Intent*

We say that the swarm has reached a consensus decision, when all robots of the swarm favor the same option. We refer to the different options of the best-of-n problem using natural numbers $1, \ldots, n$. A collective decision is represented by the establishment of a large majority $M >= (1-\delta)N$ of robots that favor the same option i, where δ verifies $0 <= \delta << 0.5$, represents a threshold set by the designer. The constraint $\delta << 0.5$ requires the opinions within the swarm to form a cohesive collective decision for a single option. In the boundary case with $\delta = 0$, we say that the swarm has reached a consensus decision. The best-of-n problem requires a swarm of robots to make a collective decision for the option n where $i \in 1, \ldots, n$ that maximizes the resulting benefits for the collective and minimizes its costs. Each option i is characterized by a quality and by a cost that are a function of one or more attributes of the target environment [31].

6.1.2 Motivation

[38] classifies instances of the best-of-n problem in five different categories depending on how the option quality and the option cost are configured in the application scenario and perceived by the robots.[38] illustrates how different combinations of option quality and option cost define the best option of the best-of-n problem. If all options have the same quality (respectively, cost), we say that the best-of-n problem is symmetric with respect to the option quality (option cost) ; If at least two options of different quality (cost) exist, we say that the best-of-n problem has asymmetric option qualities (costs); When both options qualities and option costs are symmetric, the options of the best-of-n problem are equivalent to each other and the objective of the swarm is to make a collective decision for any of them (symmetry-breaking problem); When the option qualities are symmetric but the option costs are not, the objective of the swarm is to make a collective decision for the option of minimum cost. In the opposite situation, i.e., asymmetric qualities but symmetric costs, the best option for the swarm corresponds to the option of maximum quality; Finally, when both option qualities and option costs are asymmetric, we further distinguish between two situations: in the first situation, the option qualities and the option costs are synergic and the best option has both maximum quality and minimum costs; in the second situation, they are antagonistic and the best option is characterized by a trade-off between quality and cost.

6.1.3 Known Uses

To this day, Blockchain has not addressed the 100% consensus process for swarm systems. However, the combination of blockchain technology and swarm robotic systems can provide an innovative solution for the 100% consensus decision using the robots as nodes in a network and encapsulating their transactions in blocks. The blockchain approach is driven by a blockchain-based smart contracts code that is executed and verified by the nodes. the blockchain can be in this case a platform to share knowledge safely and record votes. To achieve 100% consensus, blockchain is used to record votes in the first step, and then smart contracts code is executed to verify conditions of absolute majority. This concept has been verified in [12] for the voting process in the paragraph below.

6.1.4 Blockchain solution: S-BAC

Atomic commitment is a set of protocols that guarantee reliability is distributed database. Atomic commitment aim is to ensuring that all the changes persist or none of the changes takes place. It's the perfect implementation of a 100% consensus pattern. Atomicity is a guarantee of trust among multiple communicating parties. Chainspace is a decentralized infrastructure, known as a distributed ledger, that helps users executes user-supplied transactions on their objects. Chainspace uses Sharded Byzantine Atomic Commit (S-BAC) [2], a distributed commit protocol, to ensure consistency, safety, liveliness and security. the figure below illustrates an example of the S-BAC protocol to commit a transaction with two inputs and one output. The Sharded Byzantine Atomic Commit (S-BAC) protocol applies Atomic commit combined with a Byzantine agreement Blockchain technique: Byzantine Fault Tolerance (BFT). Given a transaction T with two inputs (o1, o2) and one output (o3), the user sends the transaction to all nodes in shards managing o1 and o2. The BFT-Initiator, a designated node in each shard, sequences the transaction T, and emits 'prepared(accept, T)' or 'prepared(abort, T)' to all nodes within the shard. Then, the BFT-Initiator of each shard assesses whether 'All proposed(accept, T)' or 'Some proposed(abort, T)' holds across shards, sequences the accept(T,*), and sends the decision to the user.

6.2 Quorum Pattern

6.2.1 Intent

The difference between the quorum/vote and the consensus mechanism is that in the quorum case, an option needs a minimum 51% rate to be agreed on as the best option, rather than the absolute majority.

6.2.2 Motivation

One on the main uses of this pattern in AI robotics context is managing byzantine robots in robotic swarms. Byzantine robots are robots that deliver a misleading, faulty and malicious behavior, due to troubled environment conditions or hacking. Tracking and excluding this type of robots is crucial. Collective decision making is a fundamental concept in this tracking/exclusion task and affect two key elements in swarm of robots: task allocation, and agreement achievement. It applies voting mechanism in its three main strategies, extensivly explained in the work [37]: (i) Direct Modulation of Voter-based Decisions (DMVD): adopt the opinion of a random neighbor; (ii)Direct Modulation of Majority-based Decisions(DMMD): adopt the opinion of the majority of the neighbors (including the robot's own opinion); (iii) Direct Comparison: adopt the opinion of a random neighbor only if the robot's current quality estimate is lower than the neighbor's quality estimate.

6.2.3 Known Uses

[12] uses blockchain to achieve agreement to shows how blockchain technology can be used in the decision making process of robotic swarms. Everytime a swarm member is in a situation requiring an agreement, it can issue a special transaction, creating an address associated with each of the possible options the robotic swarm has to choose from. After being included in a block, the information is publicly available and other swarm members can vote according to their situation by, for example, transferring one token to the address corresponding to their chosen option Agreements(e.g., by the majority rule) can be obtained rapidly and in a secure and auditable way since all robots can monitor the balance of addresses involved in the voting process.

6.2.4 Blockchain solution

Blockchain based smart contracts are proven to be useful in managing byzantine robots [35]. In a classic best of n problem scenario, robot swarm study a collective decision scenario in which robots recognize which of two features in an environment is the most frequent (in this case n=2). The Ethereum smart contract created for this matter [1] provides three functions: registerRobot, applyStrategy, and vote. Each robot sends a transaction to the registerRobot function of the smart contract. This tells the blockchain the robot's public key. The robots listen to the events created by the registerRobot function, which include the robots'initial opinions as well as the block numbers and corresponding block hashes, where the opinions are saved. The robots send transactions to the function vote . the function applyStrategy applies the decision-making strategy (DMVD,DMMD, or DC, depending on the experiment). The applyStrategy function stores the chosen opinion and block number of the stable block together with the public key of the robot in the blockchain. robotsOpinionPerBlock stores the opinion of all robots for all stable block. It allows to detect Byzantine robots.

6.3 Negotiation Pattern

6.3.1 Intent

Negotiation in artificial swarms is a decision making process that is similar to voting but requires real-time synchrony, adjustment and adaptation as the decision evolves. [32] describes the benefits of negotiation in a swarm of humans.

6.3.2 Motivation

In [17], negotiation in multi-agent systems can be viewed as a distributed search through a space of potential agreements.The dimension and topology of this space is determined by the structure of the negotiation object. when new issues are added, or old ones removed, during the course of a negotiation, then extra dimensions are added, or removed, and the number of points of agreement may increase or decrease. During the negotiation process, the participants' agreement spaces, as well as their rating functions, may change: they may expand, contract, or shift, for example because their environment changes or because they change their views. The search terminates when the required number of participants find a mutually acceptable point in the agreement space or when the protocol dictates that the search should be terminated without making an agreement [7].

6.3.3 Known Uses

[12] demonstrates that the combination peer-to-peer networks and blockchain, applied on a group of agents can help reach consensus and agreement, record the agreement without a third controlling authority. this can provide the necessary capabilities to make distributed entities operations in any multi-agent system (e.g., swarm robotics [12]) more secure, autonomous, flexible and even profitable.

6.3.4 Blockchain solution: Bazaar-blockchain [28]

Bilateral, multi-round negotiations known as Bazaar negotiations are an approach for trading Cloud services on future Cloud markets. in this context, negotiations are an exchange of offers between untrusted parties, the market participants. in order to ensure transparency and integrity during such negotiations , Blockchain-based adaption of Bazaar-negotiations on the Cloud market is used. for this matter, blockchain is implemented for reading and writing offers which are exchanged during negotiations as well as for agreements which result from such negotiations. The negotiation is stored in a tamper-safe log which allows the automatic execution and observation of the agreements with Smart Contracts.

7. LEARNING FLOWS PATTERNS - INTELLIGENCE CONTINUUM

7.1 Chain Of Responsibility Pattern (a.k.a. Pipes & Filters)

7.1.1 Intent

The chain of responsibility pattern aims to decouple a caller from its target object and creates a chain of receiver objects for a task. Each receiver contains a reference to another receiver to whom it passes a task if it cannot handle it itself.

7.1.2 Motivation

Filters and pipes are one of the most important applications of this pattern. the purpose of this application is that the data passes through a sequence of transformations (filters) connected by channels (pipes). Here the transformations do not depend on each other, the topology of the model may change as stages may be replaced, added or deleted. [39] explains how filters let an application insert its own code to intercept a request or reply at certain points along its trip from the caller to its target object and back.

7.1.3 Known Uses

the supply chain is a direct incarnation of the chain of responsibility pattern. A supply chain is a processing mechanism that involves a series of steps required to get a product or service to a user. the steps allow transforming raw materials into finished products, transporting those products, and distributing them to the end user. The entities involved in the supply chain include producers, vendors, warehouses, transportation companies, distribution centers, and retailers. The elements of a supply chain include all the functions that start with receiving an order to meeting the customer's request. These functions include product development, marketing, operations, distribution, finance, and customer service.

7.1.4 Blockchain solution

Blockchain could be useful in tracking goods and the flow of materials during the production and assembly phases [27] . Indeed, the different machines or departments that an item has to pass through towards the production or assembly site could be recorded in the blockchain. when an item travels from one station to another, the movement is integrated in the blockchain and it is possible to track the routing of each item. Thanks to the time stamp, it is possible to date back to the batch of production of the product or the moment in which it finishes a manufacturing operation, to study more carefully the characteristics of each item; moreover, the blockchain could be used also in coordination with tools of statistical process control to improve the quality control within the company, facilitating the detection of defective lots during withdrawal or recall processes. The operation of filling the transaction in the blockchain would be pursued by the operator in line which has to make the transaction before the piece is moved to the next station.

7.2 Haze cascading Pattern

7.2.1 Intent

Haze Architecture and Cascading Analytics governs continuum dynamics between IoT front-end and IA/Big Data back-end. It incarnates by a DIKW (Data Information Knowledge Wisdom) discovery pattern crossing the architecture from Device then Fog/Edge to the cloud, and a Learning Feed back loop that Feeds forward insight to Adjust either Fog/Edge or Device Algorithms. IoT-Big Data integration implies bi-directional communication. One can acquire data from sensors (e.g.monitor and control IoT devices), and also send instructions to those devices (e.g. reset them or switch them off-securely and reliably) [10].

7.2.2 Motivation

Haze cascading architecture offers fluidity and dynamism, and this will guarantee scalability to the architecture.For example, certain applications may not require cloud infrastructure at the wake of its development, but still design using restricted amount of cloud resources and connectivity, in order to be able to turn on cloud on demand of the application, without an architecture redesign.

7.2.3 Known Uses

[16] IoT-BPM architecture for smart incident management is composed of three main layers: i) The Edge/Fog based IoT Layer which ensures incident data acquisition and filtering. ii) The service oriented architecture (SOA) Layer which ensures the integration between IoT layer (service consumer) and BPM layer (service provider). This provider/-consumer paradigm is proposed to manage IoT generated events, and provide a knowledge update through a learning feedback loop. iii) The BPM Layer which is enriched, offline, by Genetic Algorithm.

7.2.4 Blockchain solution: Smart contracts Implementation in a microservices-based system

Micro-services-based systems are usually built with three layers: The graphical user interface, the API-Gateway and the micro-services. Any device can be connected to the system through the API-Gateway. The role of the API Gateway is to provide a custom interface for each type of device, usually exposing a set of REST APIs. The API-Gateway then calls the different micro-services and returns the result to the caller. The work in [36] presents the implementation of a paradigmatic example of micro-service pattern with Smart Contracts written in Solidity where variables and functions do the same tasks of the corresponding set of Java classes. The study demonstrates the feasibility of the approach and shows that it is possible to implement a simple micro-services-based system with smart contracts maintaining the same set of functionalities and results. Similarly to the common micro-service architecture, the corresponding smart contracts architecture is built on two layers: The first layer is the interface between external applications and the blockchain. It provides the Application Binary Interface (ABI) to interact with the smart contracts and the User Interface in which ABI are embedded. Specifically, an external App can use ABI exposed by each smart contract to compose and perform service requests. The ABIs allow to specify the functions to call and manage the correct format of data exchanged with the smart contracts.

8. CONCLUSION & FUTURE WORK

Based on a literature review, our proposal represents a pattern catalogue to keep the integration architect mission easier. Our future work consists of completing building the catalogue according to the guidelines described in section 4, in a way that answers the research questions rigorously. A deeper study of each of the research questions will be conducted in order to set the use cases and scenarios that will allow the evaluation and validation of the patterns proposed. Our IoT world is growing at a breathtaking pace, from 2 billion objects in 2006 to a projected 200 billion by 2020, and with 26 smart objects for every human being on Earth. To control data explosion resulting,IoT and Big data analytics processes have to be combined. On the other hand, blockchain is a disruptive technology which has revolutionized the business processes with its application and adaptability.The applicability and innovation of blockchain technology are not only limited in a specific sector. It is being continuously adapted and innovated to integrate with the bleeding-edge technologies. For some obvious recurrent paradigms, it was easy to find multiple blockchain applications implementing the paradigm. However, there are many standardization challenges to address and features to enhance to increase the usability and effectiveness of blockchain in these applications.

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